Warning: This task has a 10-minute time limit for submitting your solution. Once you request an input, the timer will start automatically, and after the deadline it will no longer be possible to submit a solution for that input. *You can always request a new input*, so don't worry if the timer expires: you will simply need to request and download a new input.

To help you with this task, we have prepared some ***solution templates****, which include only the parts for reading the input and writing the output (from keyboard and to screen). You can choose to read/write from files by uncommenting the appropriate lines of code.*

- *Download the C template: passeggiata.c*
- *Download the C++ template: passeggiata.cpp*
- *Download the C# template: passeggiata.cs*
- *Download the Go template: passeggiata.go*
- *Download the JavaScript template: passeggiata.html*
- *Download the Java template: passeggiata.java*
- *Download the Pascal template: passeggiata.pas*
- *Download the Python template: passeggiata.py*
- *Download the Visual Basic template: passeggiata.vb*

Problem description

On a cool spring day, Dario decided to take a walk through the park. In front of him there are N road segments, each of which is L_i steps long.

To take a step, Dario consumes one sandwich, so to traverse the i-th road segment Dario consumes L_i sandwiches.



At the beginning of the i-th segment there is a kiosk that sells sandwiches at a price of P_i coins each. Dario's backpack has infinite capacity and each kiosk

has an unlimited number of sandwiches available, so Dario can buy as many sandwiches as he wants at each kiosk.

What is the minimum number of coins Dario spends to reach the end of the N segments?

Input format

The first line of the input file contains an integer T, the number of test cases. Then follow T test cases, numbered from 1 to T. Each test case is preceded by a blank line.

Each test case is composed as follows:

- one line containing the integer N.
- one line containing the N integers $L_0, \ldots, L_{N-1}$.
- one line containing the N integers $P_0, \ldots, P_{N-1}$.

Output format

The output file must contain the answers to the test cases you managed to solve. For each test case you solved, the output file must contain a line with the text "`Case #test:`", where `test` is the number of the test case (starting from 1), followed by the integer `ans`.

Constraints

- $T = 20$, the input files you will download will contain exactly 20 test cases.
- $1 \leq N \leq 10^5$.
- $0 \leq L_i \leq 10^6$.
- $0 \leq P_i \leq 10^6$.

Test case groups

The test cases are divided into groups.
Each group is worth 5 points, and to obtain the score for a group it is necessary to solve correctly all the test cases belonging to it.

Some groups have additional assumptions:

`Assumptions`	`Groups`
All P_i have the same value.	1
$P_i \leq P_{i+1}$ for every $i = 0 \ldots N-2$	2, 3
$P_i \geq P_{i+1}$ for every $i = 0 \ldots N-2$	4
$N \leq 1000$	5, 6, 7
No additional assumptions	8, 9, 10

Input/output examples

Input:

```
2

3
3 2 5
4 6 2

5
2 4 6 3 1
7 3 4 2 8
```

Output:

```
Case #1: 30
Case #2: 52
```

Explanation

In the **first sample case**, Dario buys: - 5 sandwiches at kiosk 0 to traverse segments 0 and 1; - 5 sandwiches at kiosk 2 to traverse segment 2. In this case Dario spends $5 \cdot 4 + 5 \cdot 2 = 30$ coins.

In the **second sample case**, Dario buys: - 2 sandwiches at kiosk 0 to traverse segment 0; - 10 sandwiches at kiosk 1 to traverse segments 1 and 2; - 4 sandwiches at kiosk 2 to traverse segments 3 and 4. In this case Dario spends $2 \cdot 7 + 10 \cdot 3 + 4 \cdot 2 = 52$ coins.